OSF Amendment 2 – Phase 2 Method: Domain-Level False-Alarm Guarantee for Unseen Drivers

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0. Registration status

This amendment registers the Phase 2 method and its confirmatory hypotheses (H5a–H5c) before any SHRP2 data are accessed, as promised in Section 8 of the original registration ("Planned amendments", item a).

- **Original registration:** OSF Registries, 4 October 2026, https://osf.io/y5zrp (embargo until 31 October 2026). Hypotheses H1–H4 and their nine tests are unchanged by this amendment.
- **Amendment 1:** size-matched H2 sensitivity analysis and cluster-bootstrap interval for the H2 secondary criterion. Amendment 1 must be filed before this amendment.
- **Data access at filing:** the author has not accessed any version of the SHRP2 bird's-eye trajectory dataset (Jiao & Calvert, 2025; DOI 10.15787/VTT1/EFYEJR). No SHRP2 count, summary or result has informed any choice below.
- **Development data used:** INTERACTION (stand-in label: moving-pair TTC2D < 1.5 s) and highD (TTC2D < 3.0 s). Their results (Section 7) are development evidence only and are not confirmatory for any hypothesis.
- **Method specification:** frozen in the project repository as `docs/phase2_method_spec_B2.md` (v1.0 approved 10 October 2026; v1.1 approved the same day) and code `src/phase2/domain_crc.py`, both committed before any test reported below was run.

1. Rationale

A false-alarm threshold fitted on pooled source data does not keep its promise in a new domain; this amendment registers a threshold whose false-alarm rate in a new unit is controlled by construction.

- **The problem (H2).** In development, the registered split-conformal threshold (target 5%) gave false-alarm rates up to 8.0% at 4 of 12 unseen INTERACTION locations, and the dress rehearsal of H2 gave a mean held-out false-alarm rate of 7.4%.
- **Why it fails.** Pooled split conformal treats every source frame as exchangeable with every target frame. Under domain shift this is false.

- **Easy repairs failed.** Label-free prior correction (EM), importance-weighted temperature scaling and an oracle prior did not close the calibration gap in development; each failed its criterion fixed before it was run.
- **The answer registered here.** Treat each driving unit (driver or trip) as one exchangeable unit and control the false-alarm rate at the unit level. The method works in federated form: each client sends one number or one curve, never data.

The theoretical step builds on conformal risk control (Angelopoulos et al., 2024), new-group conformal inference (Dunn et al., 2023; Lee et al., 2023) and the Neyman–Pearson umbrella algorithm (Tong et al., 2018). Duchi et al. (2024) name risk control across environments as future work. The contribution claimed is the federated, unit-level false-alarm guarantee for an unseen client in crash-risk forecasting, its evaluation, and its operational cost.

2. Frozen method definitions

All definitions below are fixed now and are those of the committed code; none will be tuned on SHRP2 data.

Score. The cumulative risk at the 3 s horizon from the registered hazard model, raw (no temperature or isotonic step), so that the threshold does not depend on a recalibrator fitted elsewhere. It lies between 0 and 1 by construction (one minus the product of the survival terms).

Negative. A prediction time (10 Hz frame) that is evaluable at 3 s and has no crash or near-crash in (t, t + 3 s], with the registered label rules (frames after the event excluded; the 5–7 s buffer before an event excluded; all frames of safe baseline trips are negatives when evaluable).

Unit and its loss. A unit is a driver if the data audit finds a driver identifier, otherwise a trip, that is one crash, near-crash or baseline record (the registered rule "driver if available, else event"; each event record is one trip). The loss of unit j at threshold λ is its false-alarm rate, the share of its negatives whose score exceeds λ:

$L_j(\lambda) = (1 / n_j) \times \Sigma_i \, 1\{ s_{ij} > \lambda \}$, the sum running over the n_j negatives i of unit j.

Threshold grid. λ is chosen from 2,802 fixed values: the logistic function of t = −14, −13.99, …, +14, plus λ = 1. The grid does not depend on any data.

HP — high-probability form (primary for H5). For each of the k calibration units, $t_j(\varepsilon)$ is the smallest grid value with $L_j \leqslant \varepsilon$. The threshold is the r-th smallest of $t_1, \ldots, t_k$:

threshold = $t_{(r)}$, with $r = \lceil (k + 1)(1 - \delta) \rceil$ and $\delta = 0.10$.

If the k calibration units and a new unit are exchangeable, the probability that the new unit's false-alarm rate exceeds ε is at most δ. This needs $k \geqslant 9$.

CRC — expectation form (secondary). The smallest grid value λ with

$\Sigma_j L_j(\lambda) / (k + 1) + 1 / (k + 1) \leqslant \alpha$, the sum running over the k calibration units,

bounds the expected false-alarm rate of a new unit by α (Angelopoulos et al., 2024, Theorem 1, with one unit as one loss). It needs $\alpha \geqslant 1/(k+1)$.

Levels. $\alpha = \varepsilon \in \{0.05, 0.10\}$, with 0.05 primary. Operational level $\varepsilon_{op} = 1/36{,}000$ per frame, that is one alarm frame per hour of driving at 10 Hz, with HP only. This is stricter than one alarm episode per hour (an alarm lasting 1 s is 10 frames); episode-level rates are exploratory. The expectation form is not used at this level, because it would need $k \geqslant 35{,}999$. A unit with fewer negatives than $1/\varepsilon$ gets the threshold with zero false alarms (the first grid value at or above its highest negative score).

Federated form. Each client sends one number t_j (HP) or one curve of 2,802 values (CRC); the server needs only the order statistic or the sum of curves, so secure aggregation can hide each client's curve. The federated result equals the centralised one exactly.

Eligibility. A unit is used only if it has at least 100 frames evaluable at 3 s (10 s of driving). The guarantee does not depend on unit size, so this minimum only removes degenerate units; it is set low because SHRP2 record lengths are unknown before access. It is decided on counts before any score is looked at.

Infeasibility. HP always yields a threshold when $k \geqslant 9$ (at worst $\lambda = 1$: never warn, recall 0, reported as such). CRC is infeasible when $\alpha < 1/(k+1)$; it then declines to set a threshold, which is reported and never replaced by another rule.

3. Design on SHRP2

The H5 analysis reuses the registered H1 model; each test unit's threshold is set on the other test units (leave one unit out), so the calibration units and the new unit are all units the model never saw in training or model selection.

1. **Model.** The registered H1 model: the risk head chosen by the registered selection rule, neighbour augmentation on, five training seeds (131–135), trained and selected on the training and validation splits exactly as registered. No new model is trained for H5.
2. **Units in the analysis.** The eligible units of the test split (n of them). Validation-split units are not used for calibration in the primary analysis, because they took part in model selection (epoch and risk-head choice), which can make their scores optimistic.
3. **Leave one unit out.** Each test unit in turn is the new client; the other $n - 1$ test units are its calibration units ($k = n - 1$). They give the per-unit thresholds t_j (HP), the loss curves (CRC) and the pooled negatives (C1). Thresholds are computed separately for each seed.

4. **Per test unit:** its false-alarm rate at its own threshold (share of its negatives above it) and alarm frames per hour (rate × 36,000).
5. **Positives and recall.** A positive is a frame evaluable at 3 s with a crash or near-crash in (t, t + 3 s] (registered labels, protocol P2). Recall is the share of test positive frames whose score exceeds the threshold of their own unit, pooled over test units.
6. **Seeds.** In every bootstrap resample the same test units are used for all five seeds, and each statistic is averaged over the seeds within the resample.
7. **Uncertainty.** Paired cluster bootstrap over test units, B = 2,000, seed 131, resampling the per-unit results computed once. The leave-one-out thresholds are not recomputed in each resample; with many units each threshold changes little when one unit is removed, and this approximation is stated.

Exchangeability. Every test unit is unseen by the model, and units enter the test split at random within strata (outcome type × lighting × surface). The n test units are therefore exchangeable within strata; across strata the guarantee holds approximately, and shares by stratum are reported (Section 5). With a driver identifier the registered split is by driver, so no driver spans two splits; how a driver's stratum is assigned is fixed in the data audit, before any model is evaluated, under the registered order of work.

4. Confirmatory hypotheses H5a–H5c

Three one-sided tests form a new family, with family-wise error controlled at 0.05 by the Holm procedure within the family. H5a and H5b use the paired cluster bootstrap of Section 3 with the registered p-value $p = (1 + \#\{\Delta^*_b \leqslant \delta_0\}) / (B + 1)$; H5c uses an exact binomial test.

Test	Claim	Levels	Statistic (seed-averaged) and test	Supported if	Development value (INTERACTION recordings / locations / highD)
H5a	Per-unit control: fewer new units exceed the 5% target with HP than with the registered split conformal (C1)	HP: ε = 0.05, δ = 0.10. C1: α = 0.05, frame level, on the pooled negatives of the k calibration units	Share of test units with FA > 0.05 under C1 minus the same under HP; bootstrap, $\delta_0 = 0$	Holm-adjusted p < 0.05	Shares: 0.436 vs 0.089; 7/12 vs 1/12; 0.373 vs 0.083
H5b	Bounded recall cost: HP's recall at 3 s is non-inferior to C1's, margin 0.15	As H5a	recall(HP) minus recall(C1); bootstrap, $\delta_0 = -0.15$	Holm-adjusted p < 0.05	Differences: −0.10; −0.11; −0.01

Test	Claim	Levels	Statistic (seed-averaged) and test	Supported if	Development value (INTERACTION recordings / locations / highD)
H5c	Operational validity: fewer than 20% of new units exceed one alarm frame per hour	HP: ε_{op} = 1/36,000, δ = 0.10	x = number of test units with FA > ε_{op}, averaged over seeds and rounded up; exact one-sided binomial test of H0: share ⩾ 0.20, p = P(Bin(n, 0.20) ⩽ x)	Holm-adjusted p < 0.05	Shares: 0.084; 1/12; 0.090

- **What the guarantee implies for H5c.** With leave-one-out thresholds, each test unit exceeds ε_{op} with probability at most δ = 0.10, so the expected share is at most 0.10. The tolerance of 0.10 above this bound allows for the finite number of test units and gives the test adequate power: if the true share is 0.10, the test at level 0.05 rejects with probability about 0.62 at n = 50, 0.88 at n = 100 and 0.99 at n = 200.
- **Why an exact binomial test for H5c.** For a share near its bound with a limited number of units, bootstrap p-values are anti-conservative. The binomial test treats the units' exceedances as independent given the thresholds (each leave-one-out threshold changes little between units); rounding the seed-averaged count up is conservative.
- **Practical relevance.** H5a is labelled practically relevant if its estimate is at least 0.10 (ten percentage points fewer units above target); otherwise "supported but small".
- **Families.** H5 tests the Phase 2 method registered here, a question distinct from H1–H4. Error is controlled at 0.05 within each family; across the registered nine-test family and this family together the family-wise error is at most 0.10. As a sensitivity analysis, Holm over all 12 tests is also reported.
- **Margins.** The H5b margin (0.15) and the H5c tolerance (0.10) were set from the development results in Section 7 and a power calculation, before any SHRP2 access, and are fixed now.
- **Reporting.** Every result is reported with its estimate, 95% interval and Holm-adjusted p-value, whatever its direction; recall is never reported without the false-alarm figures, and vice versa.

5. Secondary and exploratory analyses

These analyses are reported in full but carry no confirmatory claim.

Secondary (pre-specified, descriptive):

1. H5a and H5b repeated at the 10% level (HP: ε = 0.10, δ = 0.10; C1: α = 0.10).
2. The expectation form (CRC) at 5% and 10%: mean false-alarm rate over test units against α, share of test units above α, recall. With k = n − 1 it needs n ≥ 20 at the 5% level.
3. **Deployment design.** Thresholds set once on the eligible validation units (k = their number) and applied to every test unit, as a deployed system would do. Reported for HP, CRC and C1 at every level, with the caveat that validation units took part in model selection, so their scores may be optimistic.
4. Published comparators on the same units, scores and levels, as implemented and tested in `src/phase2/comparators.py`: C2 federated conformal (Lu et al., 2023; exact quantile), C3 one-shot quantile of quantiles (Humbert et al., 2023; exact coverage for unequal client sizes), C4 risk-curve shrinkage for an unseen site (Shahid, 2026), C5 label-shift weighted conformal, class-0 form (Plassier et al., 2023), and C6, the oracle threshold of each test unit (reference only).
5. Shares of test units above target by stratum (outcome type, lighting, surface).
6. The registered pre-specified sensitivity analyses, where they apply to H5: crash-only outcome, both risk heads, and raw values without implausible-value handling.
7. Holm over all 12 confirmatory tests (the nine registered tests and H5a–H5c together), as stated in Section 4.

Exploratory (labelled as such):

1. **Environment level.** The registered H3 clients (lighting × surface × traffic density cells, at most 8) as held-out domains. With k ⩽ 7 calibration cells, HP needs δ ⩾ 1/8 and CRC needs α ⩾ 1/8, so both are infeasible at the registered levels and are reported as such; per-cell false-alarm rates of the unit-level thresholds are reported descriptively.
2. **Learning curve (drivers only).** If units are drivers, each test driver′s trips arrive in order of trip start time (trip identifier if no time is available). After the first m trips, a local HP threshold is set from 60 s segments of those trips (segment eligible with at least 200 evaluable frames) and evaluated on the driver′s remaining trips. Read-outs as in development: the first m at which the local threshold keeps ε and matches the borrowed threshold′s recall (at least 5 remaining positives), and the first m at which a temperature re-fitted on the burn-in gives a lower adaptive ECE at 3 s. Full curves are reported, not only first crossings.
3. False alarms per hour of safe baseline driving for every method and level, counted both as alarm frames and as alarm episodes (a run of consecutive alarm frames counts as one episode).

6. Contingencies and not-testable rules

If a test cannot be run under these rules it is reported as "not testable", never replaced by another test.

- **No driver identifier:** units are trips (Section 2); the exploratory learning curve is then not run.
- **Too few test units:** fewer than 20 eligible test units makes H5a, H5b and H5c not testable (too few clusters for the bootstrap; HP also needs $k = n - 1 \geqslant 9$). Fewer than 50 makes H5c not testable, because the exact binomial test then has too little power (below about 0.6 if the true share is 0.10).
- **Too few validation units:** fewer than 9 eligible validation units means the secondary deployment design (Section 5) is not run; the confirmatory tests are unaffected.
- **Too few test events:** fewer than 50 test events (crashes or near-crashes with at least one positive frame at 3 s) makes H5b not testable. This is the registered small-domain threshold, which counts events, not frames.
- **Never-warn threshold:** if HP sets $\lambda = 1$ for a test unit, that unit has no false alarms and no recall; it is included as it is and counted.
- **Primary horizon:** if 3 s fails the registered retention rule, the registered fallback horizon is used for H5 as well and reported as a deviation.
- **Bootstrap:** resamples with an undefined statistic (for example no positives) are discarded and counted; more than 5% discarded flags the test as unreliable, but it is still reported (registered rule).
- **Training:** the registered divergence rule applies; no budget is extended after seeing results.
- **Counts reported:** units excluded by eligibility, units with zero negatives, and units below $1/\varepsilon$ negatives (zero-false-alarm threshold) are counted for every level.

7. Development evidence (not confirmatory)

On INTERACTION and highD the per-unit guarantee held in every setting where its assumption holds, while the pooled method exceeded its target at a third to over half of new units; these results motivated H5 and are not evidence for it.

Analysis	Units (k calibration units)	Main result
E1 new recordings, INTERACTION	45 recordings, leave one out (k = 44); 5 seeds	Units above 5%: C1 43.6%, CRC 20.4%, HP 8.9%. Recall at 3 s: 0.55, 0.49, 0.45. Pre-set validity rule passed 20/20

Analysis	Units (k calibration units)	Main result
E1c operational level	as E1	HP: 8.4% of recordings with any false alarm; 0.59 false alarms per vehicle-hour; recall 5.9% (C1: 12.0%, 0.94, 7.5%)
E2 new locations, INTERACTION	12 locations, symmetric leave one out (k = 11); seed 131	Locations above 5%: C1 7/12 (pooled FA 5.3%), HP 1/12. Recall 0.33 vs 0.22. At 1 false alarm per hour: recall below 1%
E3a within highD	60 recordings, leave one out (k = 59); 5 seeds	Units above 5%: C1 37.3%, HP 8.3%; validity rules passed (CRC 20/20, HP 30/30)
E3b INTERACTION → highD	thresholds carried across datasets	C1 false-alarm rate 26% at a 5% target: carried thresholds break
B6 learning curve	12 locations, recordings in order	A local threshold beat the borrowed one after a median of about 10–22 minutes of recording at 9–11 of 12 locations
B7 comparators	settings A, B, C	Lu, Shahid and Plassier behave like C1. Humbert: 37.6% false alarms at a 5% target (valid, 4.7%, when clients are shuffled to be alike). Lu and Humbert cannot reach 1 false alarm per hour with 44–59 clients

Limits of this evidence: stand-in labels (TTC2D), not crashes; drone recordings of about 13 minutes; location-level results from one seed and 12 locations; the highD label threshold (3.0 s) differs from the training label (1.5 s).

8. Process notes, confirmed decisions, references

Development process notes (recorded in the project's deviations log; none involves SHRP2 data):

- P1: the method specification was committed shortly after the code for its unit tests, not before; only code and synthetic tests were committed in between.

- P2: the specification moved from v1.0 to v1.1 after the first development result (E1), making the high-probability form co-primary and adding the operational level. The change was made on development data only, before the location-level, highD and SHRP2 analyses.
- The diagnostic for comparator C3 first used a single shuffle with a rule that was too strict for a high-variance method; it was changed to 20 shuffles before it was run on real data.
- An independent statistical review of this draft on 11 October 2026, before filing and before any SHRP2 access, led to four changes: leave one test unit out as the primary design (validation units took part in model selection), an exact binomial test for H5c, frame-level definitions of positives and recall, and clearer statements of the levels and of when each form is feasible. After a power calculation, the H5c tolerance was then widened from 0.05 to 0.10: with 0.05, power was only about 0.25 at n = 50 and 0.65 at n = 200 if the true share equals the guaranteed bound of 0.10.

Decisions confirmed by the author before filing (11 October 2026):

- H5b non-inferiority margin of 0.15 (development recall cost: 0.10–0.11 on INTERACTION, 0.01 on highD)
- H5c tolerance of 0.10 above δ (that is, fewer than 20% of new units above one alarm frame per hour; widened from 0.05 for adequate power)
- H5c by an exact one-sided binomial test, with at least 50 test units
- Leave one test unit out as the primary design; validation-calibrated deployment design secondary
- H5a–H5c as a separate Holm family, beside the unchanged nine-test family
- Unit eligibility of at least 100 evaluable frames (10 s of driving)
- Primary level 5%, with 10% secondary

Use of AI assistance. As stated in the original registration, code development, literature verification and drafting were assisted by an AI system; all analyses are run, checked and approved by the author.

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